Trainings

Preparatory Steps

Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

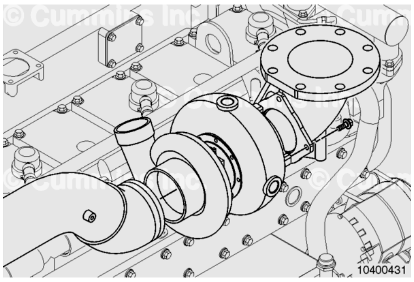
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)
- Remove the air crossover. Refer to Procedure 010-019 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html)
- Disconnect oil drain hose. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html)
- Disconnect oil supply hose. Refer to Procedure 010-046 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html)

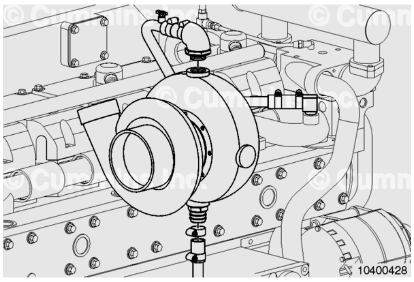
Remove

Marine Applications

Remove the intake and exhaust pipes from the turbocharger.



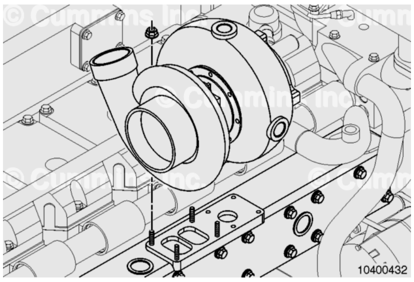
Disconnect the coolant supply hose and the coolant return hose from the turbocharger.



Remove and discard the mounting nuts. The mounting nuts can **not** be reused.

Remove the turbocharger.

Remove and discard the gasket.



Clean and Inspect for Reuse

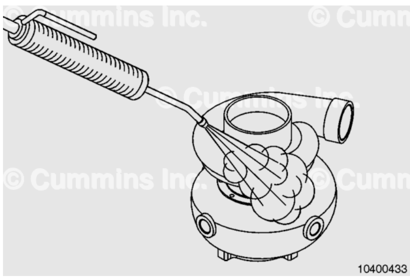
Marine Applications

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

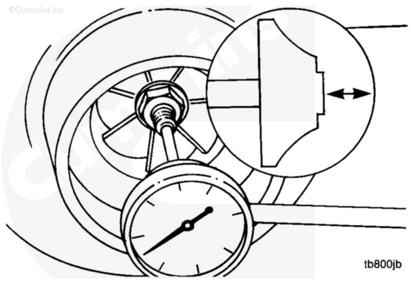
To reduce the possibility of turbocharger damage, tape or plug all openings to prevent solvent or steam from entering the oil cavities of the turbocharger.



Check for any oil leaks on the turbine or compressor seals. Refer to Procedure 010-040 (/qs3/pubsys2/xml/en/procedures/18/18-010-040.html) in Section 10 for compressor seal leaks. Refer to Procedure 010-049 (/qs3/pubsys2/xml/en/procedures/18/18-010-049.html)in Section 10 for turbine seal leaks.

Use steam to clean the turbocharger.

Check the exterior of the turbocharger for damage.



Check the wheel for fretting and broken vanes.

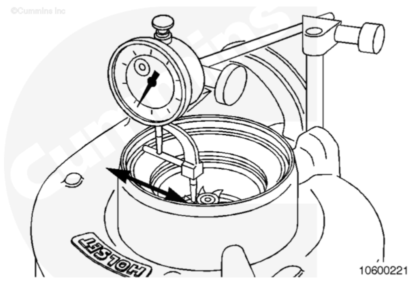
Use a dial indicator to measure the turbocharger end clearance.

Measure the end clearance.

Holset® HX82 Turbocharger End Clearance		
mm		in
0.025	MIN	0.001
0.152	MAX	0.006

If the end clearance exceeds the specifications, the turbocharger **must** be replaced.

Measure the radial clearance (side to side) at compressor impeller nose using a dial gauge



Holset® HX82 Turbocharger			
	mm		in
Compressor/Impeller	0.254	MIN	0.010
	0.787	MAX	0.031

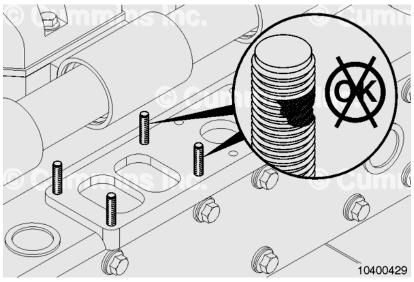
Make sure movement is within MIN/MAX Total Indicator Reading (TIR) values shown above.

If the end clearance exceeds the specifications, the turbocharger **must** be replaced.

Check the condition of the turbocharger mounting studs. Replace the studs if they are broken, corroded, worn, or if any of the threads are damaged.

Tighten the turbocharger mounting studs to verify they are tightened to the correct specification.

Torque Value: 34 n•m [25 ft-lb]

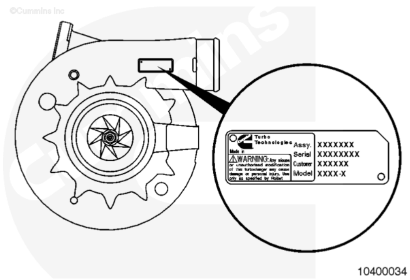


Install

Marine Applications

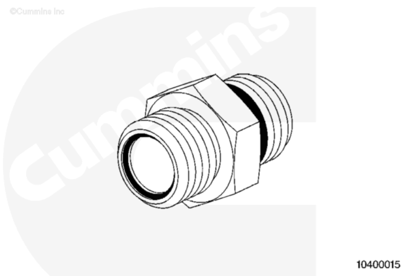
Reference the data tag on the turbocharger to determine the model.

The name is also cast on the housing of the turbocharger.



All turbocharger fittings are the flat face o-ring type fittings. Be sure the o-ring is in place before attaching the hose(s) to the fittings.

The Holset® turbocharger uses a 9/16-18 UNF, straight thread, o-ring type oil supply fitting.



Apply a heavy duty anti-seize compound to the mounting studs.

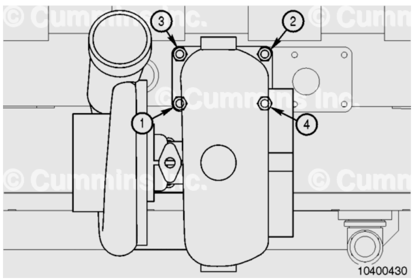
Install the gasket on the exhaust manifold with the bead up.

Install the turbocharger.

Install new turbocharger self-locking mounting nuts. Finger tighten all of the mounting nuts.

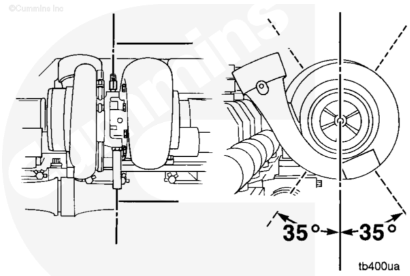
Tighten the mounting nuts in the sequence shown.

Torque Value: 57 n•m [42 ft-lb]



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



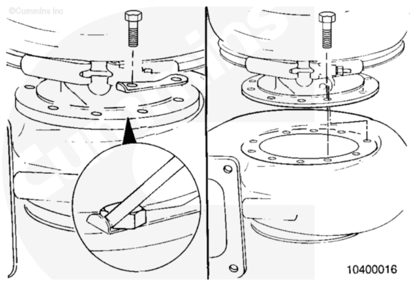
Adjust

Marine Applications

To adjust the bearing housing, bend the lockplate off the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

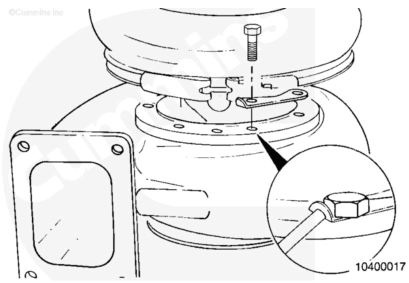


Install the lockplates and capscrews.

Tighten the capscrews.

Torque Value: 20 n•m [180 in-lb]

Bend the lockplate tabs over the capscrews.



To adjust the turbocharger compressor housing alignment:

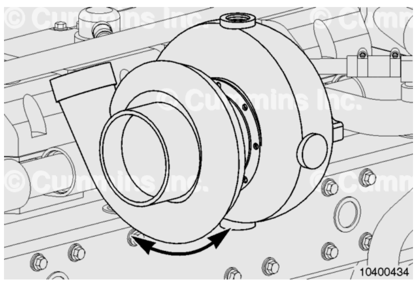
Loosen the v-band clamp.

Turn the compressor housing to the correct alignment.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

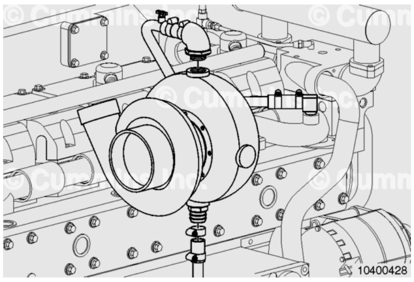
Tap the clamp with a mallet and tighten the clamp again.



Connect the coolant supply hose and the coolant return hose to the turbocharger.

Install the elbow and tighten the v-clamps.

Torque Value: 8 n•m [71 in-lb]



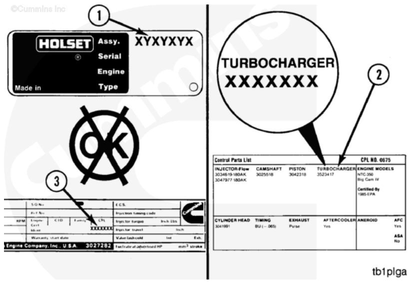
Check for Correct Component

Marine Applications

Compare the assembly number (1) on the turbocharger dataplate with the turbocharger specified in the engine Control Parts List (CPL) Manual (2).

The CPL number for each engine is listed on the engine dataplate (3).

If an incorrect turbocharger was installed, remove it and install the correct turbocharger.



Finishing Steps

Marine Applications

- Add engine oil to the turbocharger bearing housing. Connect the oil supply hose. Refer to Refer to Procedure 010-046 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-046-tr.html)
- Connect the oil drain hose. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html)
- Install the air crossover. Refer to Procedure 010-019 in Section 10. (/qs3/pubsys2/xml/en/procedures/18/18-010-019-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)

- Operate the engine until the coolant temperature is 71°C [160°F] and check for leaks.

Last Modified: 10-Jun-2021